

```
(%i3) /* Symbols */
declare([x, x_k, h], real)$;
assume(h > 0)$;
depends(f, x)$;
```

```
(%i4) /* Expanding f(x) at (x_k + h) / 2, which is the midpoint between x_k and x_{k+1} */
series: taylor(f(x), x, (x_k + h) / 2, 5);
```

series

$$f\left(\frac{x_k + h}{2}\right) + \left(\frac{d}{dx} f(x)\right)\bigg|_{x=\frac{x_k + h}{2}} \left(x - \frac{x_k + h}{2}\right) + \frac{\left(\frac{d^2}{dx^2} f(x)\right)\bigg|_{x=\frac{x_k + h}{2}} \left(x - \frac{x_k + h}{2}\right)^2}{2} +$$

$$\frac{\left(\frac{d^3}{dx^3} f(x)\right)\bigg|_{x=\frac{x_k + h}{2}} \left(x - \frac{x_k + h}{2}\right)^3}{6} + \frac{\left(\frac{d^4}{dx^4} f(x)\right)\bigg|_{x=\frac{x_k + h}{2}} \left(x - \frac{x_k + h}{2}\right)^4}{24} +$$

$$\frac{\left(\frac{d^5}{dx^5} f(x)\right)\bigg|_{x=\frac{x_k + h}{2}} \left(x - \frac{x_k + h}{2}\right)^5}{120} + \dots$$

```
(%i5) /* Integral from x_k to x_{k+1} of the series above */
expand(integrate(series, x, x_k, x_k + h));
```

(%o5)

$$\frac{h \left(\frac{d^5}{dx^5} f(x)\right)\bigg|_{x=\frac{x_k}{2} + \frac{h}{2}} x_k^5}{3840} + \frac{h \left(\frac{d^4}{dx^4} f(x)\right)\bigg|_{x=\frac{x_k}{2} + \frac{h}{2}} x_k^4}{384} + \frac{h^3 \left(\frac{d^5}{dx^5} f(x)\right)\bigg|_{x=\frac{x_k}{2} + \frac{h}{2}} x_k^3}{1152} +$$

$$\frac{h \left(\frac{d^3}{dx^3} f(x)\right)\bigg|_{x=\frac{x_k}{2} + \frac{h}{2}} x_k^3}{48} + \frac{h^3 \left(\frac{d^4}{dx^4} f(x)\right)\bigg|_{x=\frac{x_k}{2} + \frac{h}{2}} x_k^2}{192} + \frac{h \left(\frac{d^2}{dx^2} f(x)\right)\bigg|_{x=\frac{x_k}{2} + \frac{h}{2}} x_k^2}{8} +$$

$$\frac{h^5 \left(\frac{d^5}{dx^5} f(x)\right)\bigg|_{x=\frac{x_k}{2} + \frac{h}{2}} x_k}{3840} + \frac{h^3 \left(\frac{d^3}{dx^3} f(x)\right)\bigg|_{x=\frac{x_k}{2} + \frac{h}{2}} x_k}{48} + \frac{h \left(\frac{d}{dx} f(x)\right)\bigg|_{x=\frac{x_k}{2} + \frac{h}{2}} x_k}{2} + h$$

$$f\left(\frac{x_k}{2} + \frac{h}{2}\right) + \frac{h^5 \left(\frac{d^4}{dx^4} f(x)\right)\bigg|_{x=\frac{x_k}{2} + \frac{h}{2}}}{1920} + \frac{h^3 \left(\frac{d^2}{dx^2} f(x)\right)\bigg|_{x=\frac{x_k}{2} + \frac{h}{2}}}{24}$$

```
(%i6) /* Sum this over k to obtain the full integral from a to b of f(x) */
map(factor, collectterms(%,
at(diff(f(x), x, 1), x = x_k / 2 + h / 2),
at(diff(f(x), x, 2), x = x_k / 2 + h / 2),
at(diff(f(x), x, 3), x = x_k / 2 + h / 2),
at(diff(f(x), x, 4), x = x_k / 2 + h / 2),
at(diff(f(x), x, 5), x = x_k / 2 + h / 2),
f(x_k / 2 + h / 2)));
```

$$\begin{aligned}
 (%o6) \quad & h f\left(\frac{x_k + h}{2}\right) + \frac{h \left(\frac{d^4}{d x^4} f(x) \right) \Big|_{x=\frac{x_k + h}{2}} (5 x_k^4 + 10 h^2 x_k^2 + h^4)}{1920} + \\
 & \frac{h \left(\frac{d^5}{d x^5} f(x) \right) \Big|_{x=\frac{x_k + h}{2}} x_k (x_k^2 + 3 h^2) (3 x_k^2 + h^2)}{11520} + \frac{h \left(\frac{d^2}{d x^2} f(x) \right) \Big|_{x=\frac{x_k + h}{2}} (3 x_k^2 + h^2)}{24} + \\
 & \frac{h \left(\frac{d^3}{d x^3} f(x) \right) \Big|_{x=\frac{x_k + h}{2}} x_k (x_k^2 + h^2)}{48} + \frac{h \left(\frac{d}{d x} f(x) \right) \Big|_{x=\frac{x_k + h}{2}} x_k}{2}
 \end{aligned}$$